Flat Parameter Fronts, q-Susceptibility, and Smooth Dynamics

New Frontier Results in the Fabius–Rvachev System

Research report prepared from the current `ProveIt` documentation corpus

Repository snapshot surveyed: 30 August 2026

Abstract

This report develops three new, mutually reinforcing directions in the Fabius–Rvachev system. The first concerns the normalized geometric-uniform family

$$X_q = (1-q)\sum_{j\geq 0} q^j U_j, \qquad 0 < q < 1,$$

whose half-base member is the Fabius distribution. We prove that its density is jointly C^∞ in the parameter and space variables, but that the exact central plateau for $q \leq 1/2$ produces a continuum of flat nonanalytic fronts. Every mixed derivative of the plateau defect vanishes on those fronts, while the defect is strictly positive immediately on the other side. In particular,

$$\partial_q^r f_q\left(\tfrac{1}{2}\right)\Big|_{q=1/2} = 2^{r+1} r! \qquad (r \geq 0),$$

yet the Taylor series does not represent the right-hand germ. This gives a parameter-space counterpart of the classical nowhere-analytic smoothness of the Fabius function.

The second direction is a linear-response or susceptibility theory at $q = 1/2$. A pathwise parameter derivative yields a canonical signed tangent measure. We derive its contractive resolvent expansion, exact CDF and density refinement equations, conditional-transport form, infinite-sinc Fourier transform, Bernoulli cumulants, Bell-polynomial moment jets, and a rational shifted-Legendre spectrum. The first several exact Legendre coefficients are computed; a tempting alternating-sign pattern is shown to fail at degree 16.

The third direction treats composition dynamics. Using the already established strict convexity of F, the exact midpoint–endpoint transfer, and the complete affine midpoint jet, we construct a unique global C^∞ diffeomorphism $\Theta : (0,1) \to \mathbb{R}$ satisfying

$$\Theta(F(x)) = 2\Theta(x), \qquad \Theta(F^{-1}(x)) = \tfrac{1}{2}\Theta(x), \qquad \Theta'(1/2) = 1.$$

The coordinate differs from $x - 1/2$ by a nonzero flat function. Consequently inverse iterates have the beyond-all-orders expansion

$$F^{-n}(x) = \frac{1}{2} + 2^{-n}\Theta(x) + \mathcal{O}_{x,N}(2^{-Nn}) \quad \text{for every } N,$$

with no intermediate algebraic correction terms. Finally, the known quadratic endpoint law is converted into a logarithmic Böttcher-type height, giving an exact doubling cocycle and a double-exponential asymptotic for forward iterates. Its quotient by $-\Theta$ is an exact one-periodic dynamical factor. The report ends with explicit conjectures, numerical checks, and a Lean formalization roadmap.

Contents

1 Scope, corpus audit, and novelty boundary

1.1 What was surveyed

The anti-duplication audit used the live documentation tree under `Analysis/FabiusFunction/docs`, with the following hierarchy.

1. The primary exposition `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`, its mathematical glossary, and the separate non-elementarity article were treated as the formalization-backed baseline.

2. The canonical `semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex` was treated as the principal index of deductions, experiments, and open obligations not yet promoted to the primary exposition.

3. The current draft manifest and the thematic consolidation directories were used as the index of later work on Thue–Morse formulae, exponent sequences and q-series, finite sinc products, atomic functions beyond the dyadic case, signed/reciprocal q-Fabius orbits, Lambert W, spectra and arithmetic, transforms, inverse and sampling problems, Legendre/Lagrange/Rvachev representations, and Fourier decay.

4. The frozen archive was inspected as provenance, not counted as a separate source of new claims. Its own README states that live byte-identical versions were pruned and that the remaining entries are historical snapshots. Re-reading every frozen duplicate as an independent article would therefore bias the duplication audit.

The live frontier README records that eleven former notebooks were consolidated into six canonical thematic syntheses plus a gap register, preserving distinct arguments, numerical evidence, warnings, and proof obligations while compressing redundant repetitions [12]. The current manifest adds the larger thematic draft volumes and recent representation reports [13]. Accordingly, the phrase *new result* below means "not found in the active primary exposition, canonical frontier volume, or current manifest under the audited concepts and equivalent terminology." It is not a claim of absolute priority over all mathematical literature.

Repository-novel axes identified by the audit. Exact searches in the active primary and canonical frontier sources found no Fabius-specific Koenigs/Schröder conjugacy, no Böttcher treatment of composition orbits, and no systematic parameter derivative/linear-response theory for the geometric-uniform family. The corpus does contain the ingredients used here: the geometric law and its smooth density, the exact plateau phase, the midpoint transfer, inverse midpoint defect identities, strict shape theorems, Bernoulli–Bell machinery, Legendre expansions, and sharp endpoint asymptotics.

1.2 Existing inputs deliberately not reproved from first principles

We use the following results from the repository as named inputs.

- The bounded Fabius function F is C^∞, strictly increasing on $[0,1]$, constant outside that interval, symmetric by $F(1-x) = 1 - F(x)$, and satisfies

$$F'(x) = 2F(2x) \quad (0 \le x \le 1/2).$$

  It is strictly convex on $[0,1/2]$, strictly concave on $[1/2,1]$, and all positive-order endpoint derivatives vanish [11].

- The exact midpoint transfer is

$$F\left(\frac{1}{2}+h\right) = \frac{1}{2} + 2h - F(h), \qquad F\left(\frac{1}{2}-h\right) = \frac{1}{2} - 2h + F(h), \quad 0 \le h \le \frac{1}{2}, \tag{1}$$

and $F^{(k)}(1/2) = 0$ for every $k \geq 2$ [11].

- For $0 < q < 1$, the normalized geometric-uniform law has a compactly supported C^∞ density f_q, a continuous CDF F_q, reflection symmetry, exact support $[0, 1]$, and the affine refinement equation stated in section 2 [11].
- At the dyadic endpoint,
$$\log F(2^{-t}) \sim -\frac{\log 2}{2}\, t^2 \qquad (t \to \infty), \tag{2}$$
while the sharp expansion contains a nonconstant periodic correction in an exact lower-Lambert phase [11, 12].

Everything labeled Proved here is derived in this report from these inputs and standard real/Fourier analysis. Conjectures are explicitly separated.

1.3 Relation to the major existing themes

The new constructions are not isolated from the established corpus. They create a set of bridges:

Existing theme	New object	Bridge
Geometric q-law	Parameter tangent measure ν	Differentiation of the affine fixed point; contractive resolvent
Exact plateau phase	Flat nonanalytic fronts	Endpoint flatness transported into parameter space
Infinite sinc product	Tangent Fourier product	One marked factor plus a logarithmic derivative sum
Bernoulli cumulants	Moment q-jets	q-integers and complete Bell polynomials
Legendre expansion	Tangent Legendre spectrum	Exact rational coefficients from moment derivatives
Midpoint–endpoint transfer	Global Schröder coordinate	Flat local linearization extended by inverse iteration
Lambert/periodic endpoint law	Dynamical height B	Doubling cocycle and a new one-periodic quotient
Thue–Morse/dyadic structure	Composition clocks	Exact multiplication by 2 in the global coordinate

2 The normalized geometric-uniform family

Let $(U_j)_{j\geq 0}$ be independent $\mathrm{Unif}[0, 1]$ random variables and, for $0 < q < 1$, define

$$c_j(q) = (1-q)q^j, \qquad X_q = \sum_{j\geq 0} c_j(q)U_j, \qquad Z_q = X_q - \frac{1}{2}. \tag{3}$$

The weights sum to one, so $X_q \in [0, 1]$ and $\mathbb{E}X_q = 1/2$. We write

$$F_q(x) = \mathbb{P}(X_q \leq x), \qquad f_q(x) = F_q'(x).$$

At $q = 1/2$, $F_{1/2} = F$ is the bounded Fabius function and $f_{1/2} = F'$ is the affine copy of Rvachev's bump.

The independent-copy decomposition

$$X_q \overset{d}{=} (1-q)U_0 + qX_q' \tag{4}$$

gives

$$F_q(x) = \int_0^1 F_q\left(\frac{x-(1-q)u}{q}\right) \mathrm{d}u, \tag{5}$$

$$f_q(x) = \frac{F_q(x/q) - F_q((x-(1-q))/q)}{1-q}. \tag{6}$$

We use the centered characteristic function

$$\Phi_q(t) = \mathbb{E}\mathrm{e}^{\mathrm{i}tZ_q} = \prod_{j\geq 0} \operatorname{sinc}\left(\frac{c_j(q)t}{2}\right), \qquad \operatorname{sinc} z = \frac{\sin z}{z}. \tag{7}$$

3 Joint smoothness and flat nonanalytic parameter fronts

3.1 Uniform Schwartz bounds in the parameter

The repository proves smoothness in x for each fixed q. The first new step is to make the parameter dependence simultaneous.

Theorem 3.1 (Joint smoothness of the geometric-uniform density)**.** *The map*

$$(q,x) \longmapsto f_q(x)$$

is C^∞ on $(0,1)\times\mathbb{R}$. More quantitatively, for every compact $K \Subset (0,1)$ and every $a,b,N \in \mathbb{N}$,

$$\sup_{q\in K}\sup_{t\in\mathbb{R}}(1+|t|)^N \left|\partial_q^a\partial_t^b\Phi_q(t)\right| < \infty. \tag{8}$$

Consequently $q \mapsto f_q$ is C^∞ as a map from $(0,1)$ into every $C^m(\mathbb{R})$, and all parameter derivatives are supported in $[0,1]$.

Proof. Fix $K = [\alpha,\beta] \Subset (0,1)$. For every derivative order r, termwise differentiation of

$$c_j(q) = q^j - q^{j+1}$$

gives a polynomial in j times a power of q. Hence

$$\sum_{j\geq 0}\sup_{q\in K}|\partial_q^r c_j(q)| < \infty. \tag{9}$$

In particular, $q \mapsto X_q$ is C^∞ in L^∞ under the common-digit coupling.

For the stronger density statement, first use the finite products $\Phi_{q,M}(t) = \prod_{j<M}\operatorname{sinc}(c_j(q)t/2)$. Every mixed derivative is a finite Leibniz sum. A differentiated factor grows at most polynomially in $|t|$, while an undifferentiated factor satisfies

$$\left|\operatorname{sinc}\left(\frac{c_j(q)t}{2}\right)\right| \leq \min\left(1, \frac{2}{c_j(q)|t|}\right).$$

Choose an arbitrarily long fixed prefix. In a term containing a parameter derivatives and b frequency derivatives, at most $a+b$ prefix factors are marked by differentiation; all remaining prefix factors contribute inverse powers of $|t|$. Since $c_j(q) \geq (1-\beta)\alpha^j$ on a fixed prefix, choosing the prefix longer than $N + 2(a+b)+2$ gives an $(1+|t|)^{-N}$ majorant. Derivatives that land beyond the fixed prefix are summable by (9). The resulting bounds are uniform in M, so the products and all derivatives converge normally, proving (8).

The Fourier transform of f_q is $\mathrm{e}^{\mathrm{i}t/2}\Phi_q(t)$. Fourier inversion, with arbitrary powers of t allowed by (8), permits differentiation in both q and x under the integral. This proves joint C^∞ regularity. Since F_q is identically zero to the left of zero and identically one to the right of one for all $q \in (0,1)$, every parameter derivative of the density vanishes off $[0,1]$. □

Remark 3.2. The theorem is a smooth, not analytic, parameter statement. The next results show that this distinction is essential, not technical.

3.2 The exact plateau and its moving boundary

For $q \le 1/2$, the head interval $(1-q)U_0$ is at least as long as the support $[0,q]$ of the tail qX'_q. The convolution therefore has a central plateau. This plateau is already present in the repository's general-base analysis. Its parameter-space consequence is new.

Proposition 3.3 (Exact plateau and edge defects). *For $0 < q \le 1/2$,*

$$f_q(x) = \frac{1}{1-q} \qquad (q \le x \le 1-q). \tag{10}$$

Fix $x \in (0, 1/2)$, and define $\Delta(q,x) = (1-q)^{-1} - f_q(x)$. Then, in the full neighborhood $0 < q < 1-x$ of the left front $q = x$,

$$\Delta(q,x) = \begin{cases} 0, & 0 < q \le x, \\ \dfrac{F_q((q-x)/q)}{1-q}, & x < q < 1-x. \end{cases} \tag{11}$$

At the meeting point $x = 1/2$,

$$\Delta\left(q, \frac{1}{2}\right) = \begin{cases} 0, & 0 < q \le 1/2, \\ \dfrac{2F_q((2q-1)/(2q))}{1-q}, & 1/2 < q < 1. \end{cases} \tag{12}$$

The right front is obtained by $x \mapsto 1-x$.

Proof. If $q \le x \le 1-q$, then $x/q \ge 1$ and $(x-(1-q))/q \le 0$. Formula (6) gives (10).

Now let $x < 1/2$. For $x < q < 1-x$, the second CDF argument in (6) is negative, while $x/q \in (0,1)$. Thus

$$f_q(x) = \frac{F_q(x/q)}{1-q} = \frac{1 - F_q(1-x/q)}{1-q},$$

which is (11). At $x = 1/2$, reflection makes the two missing edge masses equal and gives (12). □

Theorem 3.4 (Flat nonanalytic fronts). *The defect $\Delta(q,x) = (1-q)^{-1} - f_q(x)$ is flat on each plateau front:*

$$\partial_q^r \partial_x^s \Delta(q,x)\big|_{q=x} = 0 \qquad (0 < x < 1/2), \tag{13}$$

$$\partial_q^r \partial_x^s \Delta(q,x)\big|_{q=1-x} = 0 \qquad (1/2 < x < 1), \tag{14}$$

for all $r, s \ge 0$, and the same statement holds at $(q,x) = (1/2, 1/2)$. Nevertheless Δ is strictly positive immediately outside the plateau. Therefore:

1. *$(q,x) \mapsto f_q(x)$ is not real analytic at any point of the two fronts;*
2. *for every fixed $x \in (0,1)$, the section $q \mapsto f_q(x)$ is not real analytic at $q_c = \min(x, 1-x)$;*
3. *the $C^\infty(\mathbb{R})$-valued map $q \mapsto f_q$ is not real analytic at any $q \in (0, 1/2]$.*

Proof. By theorem 3.1, Δ is jointly C^∞. On one open side of each front it is identically zero by (10). Every mixed derivative is therefore zero on that side and, by continuity, on the boundary. This proves flatness.

For $x < 1/2$ and $q > x$ sufficiently close to x, formula (11) has a positive argument. The support of X_q is the entire interval $[0,1]$, and a direct finite-digit event also shows $F_q(y) > 0$ for every $y > 0$; hence the defect is strictly positive. The center formula is similarly positive for $q > 1/2$.

A real-analytic function whose full Taylor series is zero at a point vanishes in a neighborhood. Thus the positive defect contradicts analyticity in one or two variables. For the function-space statement, compose a hypothetical analytic map $q \mapsto f_q \in C^\infty$ with the continuous linear evaluation functional at $x = q_0$ when $q_0 < 1/2$, or at $x = 1/2$ when $q_0 = 1/2$. □

Corollary 3.5 (Exact front jets)**.** *For every $r \geq 0$,*

$$\partial_q^r f_q(x)\big|_{q=x} = \frac{r!}{(1-x)^{r+1}}, \qquad 0 < x < 1/2, \tag{15}$$

$$\partial_q^r f_q(x)\big|_{q=1-x} = \frac{r!}{x^{r+1}}, \qquad 1/2 < x < 1, \tag{16}$$

$$\partial_q^r f_q\left(\frac{1}{2}\right)\bigg|_{q=1/2} = 2^{r+1} r!. \tag{17}$$

Every mixed jet containing at least one x-derivative vanishes on the front.

Proof. Subtract the flat defect from $(1-q)^{-1}$ and differentiate. The last statement follows because the plateau branch has no x-dependence. □

Remark 3.6 (A smooth phase transition). At $(1/2, 1/2)$, the Taylor series in $q - 1/2$ is exactly the convergent Taylor series of $(1-q)^{-1}$. It therefore predicts the plateau continuation on both sides, while the true right-hand function is smaller by the positive, flat term in (12). This is a particularly transparent example of a convergent Taylor series that fails to represent its C^∞ function.

4 The first q-response at the Fabius point

4.1 Pathwise differentiation and the tangent measure

Differentiate the weights in the common-digit coupling. At $q = 1/2$,

$$d_j := c_j'(1/2) = \begin{cases} -1, & j = 0, \\ 0, & j = 1, \\ (j-1)2^{-j}, & j \geq 2. \end{cases} \qquad \sum_{j \geq 0} d_j = 0. \tag{18}$$

Define

$$X = X_{1/2}, \qquad Z = X - \frac{1}{2}, \qquad V = \sum_{j \geq 0} d_j U_j = \sum_{j \geq 0} d_j \left(U_j - \frac{1}{2}\right). \tag{19}$$

Theorem 4.1 (Pathwise linear response)**.** *For every $\varphi \in C^1(\mathbb{R})$ whose derivative is bounded,*

$$\frac{\mathrm{d}}{\mathrm{d}q} \mathbb{E}\varphi(X_q)\bigg|_{q=1/2} = \mathbb{E}\big[\varphi'(X)V\big]. \tag{20}$$

There is a unique compactly supported smooth signed density

$$\rho(x) = \partial_q f_q(x)\big|_{q=1/2} \tag{21}$$

representing this functional, and its cumulative response

$$G(x) = \partial_q F_q(x)\big|_{q=1/2} = \int_{-\infty}^{x} \rho(t)\,\mathrm{d}t \tag{22}$$

satisfies

$$G(1-x) = -G(x), \qquad \rho(1-x) = \rho(x), \tag{23}$$

$$\int \rho = 0, \qquad \int \left(x - \frac{1}{2}\right) \rho(x)\,\mathrm{d}x = 0, \qquad \rho\left(\frac{1}{2}\right) = 4. \tag{24}$$

Proof. The derivative series $\sum c_j'(q) U_j$ converges uniformly for q in a compact subinterval of $(0, 1)$, so the ordinary chain rule and dominated convergence give (20). Joint smoothness from theorem 3.1 supplies ρ and permits differentiation of the CDF.

Replacing every digit by $1 - U_j$ sends X to $1 - X$ and, because $\sum d_j = 0$, sends V to $-V$. This gives (23). Total mass and mean are independent of q, proving the two integral identities. The center value is the case $r = 1$ of (17). □

4.2 A contractive susceptibility resolvent

The response is not only a derivative of an infinite product; it is the solution of a contractive fixed-point equation.

Let $\mathcal{M}_0$ be the space of finite signed measures of total mass zero and finite first moment, equipped with the Kantorovich–Rubinstein norm

$$\|\eta\|_{\mathrm{KR}} = \sup\left\{\left|\int \phi \,\mathrm{d}\eta\right| : \phi(0) = 0,\ \mathrm{Lip}(\phi) \le 1\right\}. \tag{25}$$

For $0 < q < 1$, define

$$\langle \phi, R_q\eta\rangle = \int_0^1 \int \phi((1-q)u + qx)\,\mathrm{d}\eta(x)\,\mathrm{d}u. \tag{26}$$

Theorem 4.2 (Susceptibility series)**.** *The operator R_q is a contraction on $\mathcal{M}_0$:*

$$\|R_q\eta\|_{\mathrm{KR}} \le q\|\eta\|_{\mathrm{KR}}. \tag{27}$$

Let μ_q be the law of X_q, and define the source functional

$$\langle \phi, \sigma_q\rangle = \int_0^1 \int \phi'((1-q)u + qx)(x-u)\,\mathrm{d}\mu_q(x)\,\mathrm{d}u. \tag{28}$$

Then the parameter derivative $\nu_q = \partial_q\mu_q$ is the unique zero-mass solution of

$$\nu_q = R_q\nu_q + \sigma_q, \tag{29}$$

and

$$\boxed{\nu_q = (I - R_q)^{-1}\sigma_q = \sum_{n=0}^{\infty} R_q^n\sigma_q} \tag{30}$$

with convergence in $\|\cdot\|_{\mathrm{KR}}$.

Proof. If $\mathrm{Lip}(\phi) \le 1$, then $x \mapsto \int_0^1 \phi((1-q)u + qx)\,\mathrm{d}u$ has Lipschitz constant at most q. This proves (27). Differentiate the fixed-point identity $\mu_q = R_q\mu_q$: the derivative hitting the measure gives $R_q\nu_q$, while the derivative hitting the affine map gives σ_q. Since $\|R_q\| < 1$, the Neumann series is convergent and the solution is unique. ☐

Corollary 4.3 (Higher response hierarchy)**.** *For $s \ge 1$, let $\nu_q^{(s)} = \partial_q^s\mu_q$ and $R_q^{(j)} = \partial_q^j R_q$ in the distributional test-function sense. Then*

$$\nu_q^{(s)} = (I - R_q)^{-1} \sum_{j=1}^{s} \binom{s}{j} R_q^{(j)} \nu_q^{(s-j)}. \tag{31}$$

Thus all q-jets are generated by one contractive resolvent and a triangular sequence of explicit sources.

Proof. Differentiate $\mu_q = R_q\mu_q$ s times, isolate the term $R_q\nu_q^{(s)}$, and apply $(I - R_q)^{-1}$. ☐

4.3 Exact refinement equations for the tangent CDF and density

Put $f = F'$. Differentiating (5) at $q = 1/2$, then changing variables from u to $v = 2x - u$, gives the following.

Theorem 4.4 (Tangent refinement system)**.** *With F, G, f, ρ extended by their natural constant or zero tails,*

$$\boxed{G(x) = \int_{2x-1}^{2x} G(v)\,\mathrm{d}v + 4\int_{2x-1}^{2x} (x-v)f(v)\,\mathrm{d}v.} \tag{32}$$

The source can be written without f as

$$4F(x) - 4xF(2x) - 4(1-x)F(2x-1). \tag{33}$$

Differentiation in x yields

$$\boxed{\rho(x) = 2\big(G(2x) - G(2x-1)\big) + 2f(x) - 8xf(2x) - 8(1-x)f(2x-1).} \tag{34}$$

Proof. For $A(q,u,x) = (x-(1-q)u)/q$, one has $\partial_q A|_{q=1/2} = 4(u-x)$. Differentiating (5) gives

$$G(x) = \int_0^1 G(2x-u)\,\mathrm{d}u + 4\int_0^1 (u-x)f(2x-u)\,\mathrm{d}u,$$

which is (32). Integration by parts in the source and the half-base CDF identity $F(x) = \int_{2x-1}^{2x} F(v)\,\mathrm{d}v$ give (33). Leibniz's rule and $F(2x) - F(2x-1) = f(x)/2$ give (34). □

Remark 4.5. At $x = 1/2$, the boundary values of f vanish and $G(0) = G(1) = 0$; hence (34) immediately recovers $\rho(1/2) = 2f(1/2) = 4$.

4.4 Conditional transport form

Proposition 4.6 (Velocity field)**.** *Let $b(x) = \mathbb{E}[V \mid X = x]$ be any regular conditional version. On $(0,1)$,*

$$b(x)f(x) = -G(x), \qquad \rho(x) = -\frac{\mathrm{d}}{\mathrm{d}x}\big(b(x)f(x)\big), \qquad b(1-x) = -b(x). \tag{35}$$

Thus the q-response is a one-dimensional continuity equation: mass is transported by the conditional parameter velocity b.

Proof. For a smooth compactly supported test function,

$$\int \varphi(x)\rho(x)\,\mathrm{d}x = \mathbb{E}[\varphi'(X)V] = \int \varphi'(x)b(x)f(x)\,\mathrm{d}x.$$

Hence $\rho = -(bf)'$ in distributions. Both sides are smooth in the interior. Integration from the left endpoint, where bf and G vanish, gives $bf = -G$. Reflection of (X,V) gives the oddness of b. □

5 Fourier, Bernoulli–Bell, and Legendre faces of the response

5.1 A marked infinite-sinc product

At $q = 1/2$, write $c_j = 2^{-j-1}$. The centered Fourier transform of the tangent density is the parameter derivative of Φ_q.

Theorem 5.1 (Tangent sinc product)**.** *For every real or complex t,*

$$\dot\Phi(t) := \partial_q \Phi_q(t)|_{q=1/2} = \sum_{j\ge 0} d_j \frac{t}{2}\operatorname{sinc}'\left(\frac{t}{2^{j+2}}\right)\prod_{k\ne j}\operatorname{sinc}\left(\frac{t}{2^{k+2}}\right), \tag{36}$$

where d_j is given by (18). Off the zero set of Φ, this is equivalently

$$\dot\Phi(t) = \Phi(t)\sum_{j\ge 0} d_j \frac{t}{2}\left[\cot\left(\frac{c_j t}{2}\right) - \frac{2}{c_j t}\right]. \tag{37}$$

The apparent poles in (37) are removable after multiplication by Φ, as certified by the entire expression (36). The noncentered Fourier transform is $\widehat{\rho}(t) = \mathrm{e}^{\mathrm{i}t/2}\dot\Phi(t)$.

Proof. Differentiate the normally convergent product (7). The logarithmic form uses $\operatorname{sinc}'(z)/\operatorname{sinc}(z) = \cot z - z^{-1}$. □

5.2 Closed cumulants and first moment jets

Let B_r denote the Bernoulli numbers with $B_1 = -1/2$, and let

$$[r]_q = \frac{1-q^r}{1-q}$$

be the q-integer.

Theorem 5.2 (Bernoulli–q-integer cumulants). *For $r \geq 2$, the centered cumulants of Z_q are*

$$\kappa_r(q) = \frac{B_r}{r}\frac{(1-q)^r}{1-q^r} = \frac{B_r}{r}\frac{(1-q)^{r-1}}{[r]_q}. \tag{38}$$

At the Fabius point,

$$\kappa_r(1/2) = \frac{B_r}{r(2^r-1)}, \tag{39}$$

$$\dot{\kappa}_r := \partial_q \kappa_r(q)|_{q=1/2} = -2B_r \frac{2^r-2}{(2^r-1)^2}. \tag{40}$$

All odd values with $r > 1$ vanish.

Proof. The centered uniform digit has cumulant generating function

$$\log \frac{\sinh(z/2)}{z/2} = \sum_{r \geq 2} \frac{B_r}{r}\frac{z^r}{r!}.$$

Cumulants add under independence and scale homogeneously, so summing $c_j(q)^r$ gives (38). Substitution and differentiation at $q = 1/2$ give the remaining formulas. □

Let $\mu_n(q) = \mathbb{E}Z_q^n$, $\mu_n = \mu_n(1/2)$, and $\dot{\mu}_n = \partial_q \mu_n(q)|_{1/2}$.

Corollary 5.3 (Bell moment response). *For every $n \geq 1$,*

$$\boxed{\dot{\mu}_n = \sum_{r=1}^{n} \binom{n}{r} \dot{\kappa}_r \mu_{n-r}.} \tag{41}$$

In particular, for even orders,

$$\dot{\mu}_{2n} = \sum_{m=1}^{n} \binom{2n}{2m} \dot{\kappa}_{2m} \mu_{2n-2m}. \tag{42}$$

The first values are

n	μ_n	$\dot{\mu}_n$
2	$1/36$	$-2/27$
4	$19/10800$	$-83/10125$
6	$583/3810240$	$-599/617400$
8	$132809/8328096000$	$-57910967/464551605000$
10	$46840699/24990950476800$	$-32576528018233/1901452086371376000$

Proof. If $M_q(z) = \mathbb{E}e^{zZ_q} = \exp K_q(z)$, then $\partial_q M_q = M_q \partial_q K_q$. Comparing exponential generating coefficients gives (41). □

Proposition 5.4 (All parameter jets and complete Bell polynomials). *For $s \geq 1$, put $K_j(z) = \partial_q^j \log M_q(z)$. Then*

$$\partial_q^s M_q(z) = M_q(z)\,\mathcal{B}_s\big(K_1(z), \dots, K_s(z)\big), \tag{43}$$

where $\mathcal{B}_s$ is the complete exponential Bell polynomial. At $q = 1/2$, every coefficient is rational. More precisely, the odd part of the denominator of $\partial_q^s \kappa_r(1/2)$ divides the odd part of

$$\operatorname{den}\left(\frac{B_r}{r}\right)(2^r - 1)^{s+1}. \tag{44}$$

Proof. Equation (43) is the Faà di Bruno formula for the exponential. The cumulant is a rational function with denominator $1 - q^r$. After s derivatives its denominator divides a power $(1 - q^r)^{s+1}$; evaluation at $q = 1/2$ introduces only powers of two in addition to $(2^r - 1)^{s+1}$. □

5.3 An exact rational shifted-Legendre spectrum

Let P_k be the ordinary Legendre polynomial and $P_k^*(x) = P_k(2x - 1)$. Orthogonality on $[0, 1]$ gives

$$\int_0^1 P_k^*(x) P_\ell^*(x)\,\mathrm{d}x = \frac{\delta_{k\ell}}{2k + 1}.$$

Since ρ is symmetric about $1/2$, only even degrees occur.

Theorem 5.5 (Legendre coefficients of the tangent density). *Write*

$$\rho(x) = \sum_{n \geq 1} a_{2n} P_{2n}(2x - 1). \tag{45}$$

The expansion converges in $C^m[0, 1]$ for every m, and the coefficients are rational. If $P_{2n}(y) = \sum_{j=0}^n p_{n,j} y^{2j}$, then

$$\boxed{a_{2n} = (4n + 1) \sum_{j=1}^{n} p_{n,j} 2^{2j} \dot{\mu}_{2j}.} \tag{46}$$

The first coefficients are

$$\begin{aligned} a_2 &= -\frac{20}{9}, & a_4 &= \frac{1088}{225}, & a_6 &= -\frac{37076}{11025}, \\ a_8 &= \frac{49280096}{75907125}, & a_{10} &= -\frac{65166997444}{756561977325}, & a_{12} &= \frac{10886281949792992}{31325448671141625}. \end{aligned} \tag{47}$$

Moreover $a_{2n} = \mathcal{O}_A(n^{-A})$ for every $A > 0$.

Proof. Orthogonality gives

$$a_k = (2k + 1) \int_0^1 \rho(x) P_k(2x - 1)\,\mathrm{d}x = (2k + 1)\,\partial_q \mathbb{E} P_k(2Z_q)\big|_{q=1/2}.$$

Expanding the polynomial proves (46). Rationality follows from theorem 5.3. The density and all its derivatives vanish at the endpoints, so repeated integration by parts through the Legendre Sturm–Liouville operator gives decay faster than every power and convergence in every C^m norm. □

Remark 5.6 (A useful failed pattern). The signs alternate through degrees $2, 4, \dots, 14$, which might suggest $(-1)^n a_{2n} > 0$. Exact arithmetic disproves this at degree 16:

$$a_{16} = -\frac{20997028161526723805395541511872}{64367687808725429829005095415625} \approx -0.3262044805.$$

Thus any eventual sign theorem must involve a more subtle phase, and the low-degree pattern should not be promoted to a conjecture without additional asymptotic analysis.

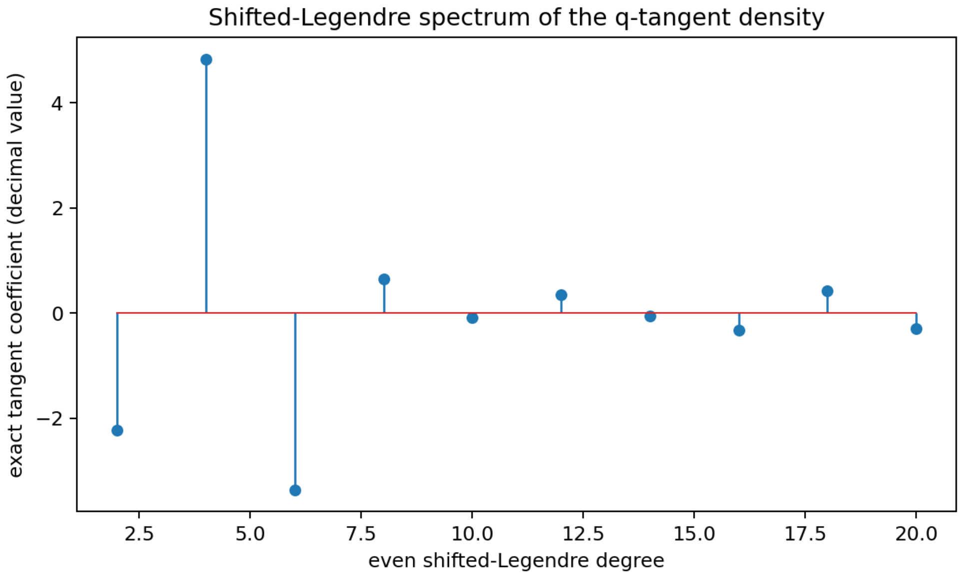

Figure 1: Exact shifted-Legendre coefficients of the tangent density through degree 20, shown in decimal form. The same-sign pair at degrees 14 and 16 is visible.

6 A global smooth Schröder coordinate for Fabius iteration

Let

$$p = \frac{1}{2}, \qquad Q = F^{-1} : (0,1) \to (0,1).$$

The primary exposition already supplies the smooth inverse, strict curvature, and the exact midpoint defect. Here we use those facts to solve the global composition dynamics.

6.1 The global basin of the midpoint

Lemma 6.1 (Diagonal inequalities and inverse attraction)**.** *For $0 < x < p$, $F(x) < x$; for $p < x < 1$, $F(x) > x$. Consequently*

$$Q^n(x) \longrightarrow p \qquad (x \in (0,1)). \tag{48}$$

Proof. On $[0,p]$, strict convexity places the graph strictly below the chord joining $(0,0)$ and (p,p), which is the diagonal. Reflection gives the right inequality. If $x < p$, then $Q(x) > x$ and $Q(x) < p$; the increasing bounded orbit converges to a fixed point. The only fixed points in $(0,p]$ are p, because the diagonal inequality excludes all others. The right half is symmetric. □

6.2 A flat local contraction

Center the inverse at p:

$$g(y) = Q(p+y) - p, \qquad |y| < p. \tag{49}$$

The midpoint transfer (1) implies, for $\delta \geq 0$,

$$g(\delta) = \frac{\delta}{2} + \frac{1}{2}F(g(\delta)). \tag{50}$$

Reflection makes g odd.

Lemma 6.2 (Flat local normal form)**.** *There is a smooth flat function r at zero such that*

$$g(y) = \frac{y}{2} + r(y), \qquad r^{(k)}(0) = 0 \quad (k \geq 0). \tag{51}$$

The function r is odd and nonzero; for $y > 0$, $r(y) > 0$.

Proof. Equation (50) gives $r(\delta) = F(g(\delta))/2$ for $\delta \geq 0$. Endpoint flatness says $F(t) = o(t^N)$ as $t \downarrow 0$ for every N, while $g(\delta) = \mathcal{O}(\delta)$. Hence r is flat from the right, and oddness gives the full statement. Positivity follows from $F(t) > 0$ for $t > 0$. □

We record the elementary smooth-linearization lemma used below.

Lemma 6.3 (Flat Koenigs lemma)**.** *Let g be C^∞ near zero, with*

$$g(0) = 0, \qquad g'(0) = \lambda \in (0,1), \qquad g(y) - \lambda y \textit{ flat at } 0.$$

Then on a neighborhood of zero the limit

$$\theta(y) = \lim_{n\to\infty} \lambda^{-n} g^n(y) \tag{52}$$

exists in C^∞, is a local increasing diffeomorphism, and satisfies

$$\theta(g(y)) = \lambda\theta(y), \qquad \theta'(0) = 1, \qquad \theta(y) - y \textit{ is flat at } 0. \tag{53}$$

It is the unique C^1 solution with derivative one at zero.

Proof. Write $g(y) = \lambda y + r(y)$. Shrink the neighborhood so that $|g(y)| \leq a|y|$ with $\lambda < a < \sqrt{\lambda}$. The telescoping identity is

$$\lambda^{-n} g^n(y) = y + \sum_{k=0}^{n-1} \lambda^{-k-1} r(g^k(y)). \tag{54}$$

Because $r(z) = \mathcal{O}(z^N)$ for every N, choose N so that $a^N < \lambda$. The series is uniformly convergent. Differentiating its terms and using the chain rule gives the same geometric majorants for every derivative order after increasing N; hence convergence is in C^∞. The functional equation follows by shifting the limit. For each fixed N, the infinite series in (54) is $\mathcal{O}(y^N)$, proving flatness. Since $\theta'(0) = 1$, it is locally invertible. If $\widetilde{\theta}$ is another normalized solution, then

$$\widetilde{\theta}(y) = \lambda^{-n}\widetilde{\theta}(g^n(y)) = \lambda^{-n} g^n(y) + o(1),$$

so it equals the limit. □

6.3 Globalization and exact conjugacy

Theorem 6.4 (Global Fabius Schröder coordinate)**.** *There is a unique C^∞ increasing diffeomorphism*

$$\Theta : (0,1) \longrightarrow \mathbb{R} \tag{55}$$

with

$$\Theta(p) = 0, \qquad \Theta'(p) = 1, \tag{56}$$

that satisfies either, hence both, of the conjugacy equations

$$\boxed{\Theta(Q(x)) = \frac{1}{2}\Theta(x), \qquad \Theta(F(x)) = 2\Theta(x).} \tag{57}$$

It is odd under reflection,

$$\Theta(1-x) = -\Theta(x), \tag{58}$$

and has the global limit representation

$$\boxed{\Theta(x) = \lim_{n\to\infty} 2^n \left(Q^n(x) - \frac{1}{2}\right).} \tag{59}$$

At the midpoint,

$$\Theta(p+y) = y + \mathrm{Flat}(y), \qquad \Theta^{-1}(t) = p + t + \mathrm{Flat}(t), \tag{60}$$

where each displayed remainder is nonzero and flat.

Proof. Apply theorem 6.3 to g with $\lambda = 1/2$, obtaining a local coordinate θ. By theorem 6.1, every interior point enters the local neighborhood under some inverse iterate. If m is large enough, define

$$\Theta(x) = 2^m \theta(Q^m(x) - p). \tag{61}$$

The local equation $\theta(g(y)) = \theta(y)/2$ shows that the value is independent of the admissible m. Locally one may choose the same m for nearby x, so Θ is C^∞. Its derivative is positive because Q^m and θ are increasing. The conjugacy and normalization are immediate, and the limit formula follows from the local limit after a finite initial segment.

The image is an open interval containing zero. It is invariant under multiplication by two because $\Theta(F(x)) = 2\Theta(x)$. Since it contains both positive and negative values, it must be all of $\mathbb{R}$. Reflection and local uniqueness give oddness. Global uniqueness follows from

$$\widetilde{\Theta}(x) = 2^n \widetilde{\Theta}(Q^n(x)) = 2^n(Q^n(x) - p) + o(1).$$

Flatness is the local conclusion of theorem 6.3; flatness of the inverse follows by composition. The remainder is not identically zero because that would make F exactly affine near p, contradicting (1) and $F(h) > 0$ for $h > 0$. □

Corollary 6.5 (Exact iterate formula and beyond-all-orders inverse asymptotic)**.** *For every $x \in (0, 1)$ and $n \geq 0$,*

$$F^n(x) = \Theta^{-1}(2^n \Theta(x)), \qquad Q^n(x) = \Theta^{-1}(2^{-n}\Theta(x)). \tag{62}$$

For every integer $N \geq 2$,

$$\boxed{Q^n(x) = \frac{1}{2} + 2^{-n}\Theta(x) + \mathcal{O}_{x,N}(2^{-Nn}) \qquad (n \to \infty).} \tag{63}$$

Thus every algebraic correction between 2^{-n} and 2^{-Nn} vanishes: the first correction is beyond all orders in the contraction scale.

Proof. Equation (62) is iteration of the conjugacy. Since $\Theta^{-1}(t) - p - t$ is flat, it is $\mathcal{O}(t^N)$ for every N. Substitute $t = 2^{-n}\Theta(x)$. □

Corollary 6.6 (Nonanalyticity of the linearizing coordinate)**.** *Neither Θ nor Θ^{-1} is real analytic at the midpoint. Their Taylor series there are exactly $x - p$ and $p + t$, respectively.*

Proof. If Θ were analytic, its flat difference from $x - p$ would vanish locally. The conjugacy would then force $F(p + h) = p + 2h$ locally, contradicting $F(p + h) = p + 2h - F(h)$. The inverse case is equivalent. □

6.4 Exact escape clocks

Corollary 6.7 (Dyadic escape time)**.** *Fix $a \in (0, 1/2)$. For $x \in (p, p + a)$, let*

$$\tau_a(x) = \min\{n \geq 0 : F^n(x) \geq p + a\}.$$

Then

$$\tau_a(x) = \left\lceil \log_2 \frac{\Theta(p + a)}{\Theta(x)} \right\rceil. \tag{64}$$

The left-half formula is obtained by absolute values. As $x \to p$,

$$\log_2 \frac{\Theta(p + a)}{|\Theta(x)|} = -\log_2 |x - p| + \log_2 \Theta(p + a) + \mathrm{Flat}(x - p). \tag{65}$$

Proof. In the Θ-coordinate, one forward iterate is multiplication by two. Therefore the first crossing of $p + a$ is the first n with $2^n\Theta(x) \geq \Theta(p + a)$. Flatness gives the final expansion. □

7 A logarithmic Böttcher height at the endpoints

The coordinate Θ solves the hyperbolic dynamics globally, but its endpoint inverse $\Theta^{-1}(t)$ as $|t| \to \infty$ is highly non-elementary. The leading endpoint law already implies a second natural doubling coordinate adapted to logarithmic height.

Put

$$L = \log 2, \qquad a = \frac{1}{2L}. \tag{66}$$

Equation (2) is equivalent to

$$-\log F(\mathrm{e}^{-s}) = as^2(1 + o(1)) \qquad (s \to \infty). \tag{67}$$

For $0 < x < p$, define

$$z(x) = \log\bigl(a[-\log x]\bigr), \qquad \varepsilon(x) = z(F(x)) - 2z(x). \tag{68}$$

Then $\varepsilon(x) \to 0$ as $x \downarrow 0$.

Theorem 7.1 (Endpoint dynamical height). *For every $x \in (0, p)$, the limit*

$$\boxed{B(x) = \lim_{n\to\infty} 2^{-n} \log\left(a[-\log F^n(x)]\right)} \tag{69}$$

exists, is positive, and is continuous in x. It has the convergent cocycle series

$$B(x) = z(x) + \sum_{k=0}^{\infty} 2^{-k-1}\varepsilon(F^k(x)) \tag{70}$$

and satisfies the exact Böttcher-type equation

$$B(F(x)) = 2B(x). \tag{71}$$

Moreover, for every fixed $x \in (0, p)$,

$$-\log F^n(x) = 2L \exp\bigl(2^n B(x) + o(1)\bigr), \tag{72}$$

or equivalently

$$\boxed{F^n(x) = \exp\left\{-2L \exp\bigl(2^n B(x) + o(1)\bigr)\right\}.} \tag{73}$$

The right-endpoint statement follows by replacing x with $1 - x$.

Proof. Let $x_n = F^n(x)$, $s_n = -\log x_n$, and $z_n = \log(as_n)$. Since $F(x) < x$ on the left half, $x_n \downarrow 0$. Equation (67) gives

$$z_{n+1} = 2z_n + \varepsilon(x_n), \qquad \varepsilon(x_n) \to 0. \tag{74}$$

Thus

$$2^{-n} z_n = z_0 + \sum_{k=0}^{n-1} 2^{-k-1}\varepsilon(x_k),$$

and the series converges because its terms are eventually bounded by a geometric sequence. This proves (69) and (70). Uniform convergence on compact subsets follows from monotone convergence of the iterates to zero and the uniform smallness of ε near zero, giving continuity. Shifting the orbit proves (71).

The tail identity

$$2^n B(x) - z_n = \sum_{k=n}^{\infty} 2^{n-k-1}\varepsilon(x_k)$$

tends to zero. Therefore $z_n = 2^n B(x) + o(1)$, which gives (72) and (73). Positivity follows because $z_n \to +\infty$; if $B(x) = 0$, the tail identity would instead force $z_n \to 0$. ☐

7.1 An exact periodic quotient of the two dynamical coordinates

Both B and $-\Theta$ double under F on the left half. Their quotient is therefore invariant along each orbit. In the linear coordinate, invariance becomes periodicity.

Theorem 7.2 (Dynamical one-periodic factor). *Define, for $u \in \mathbb{R}$,*

$$\mathcal{P}_{\text{dyn}}(u) = 2^{-u} B\big(\Theta^{-1}(-2^u)\big) . \tag{75}$$

Then $\mathcal{P}_{\text{dyn}}$ is positive, continuous, and one-periodic. For every $x \in (0, p)$,

$$\boxed{B(x) = (-\Theta(x))\, \mathcal{P}_{\text{dyn}}(\log_2[-\Theta(x)]) .} \tag{76}$$

Proof. If $x_u = \Theta^{-1}(-2^u)$, then $x_{u+1} = F(x_u)$ by theorem 6.4. Hence

$$\mathcal{P}_{\text{dyn}}(u+1) = 2^{-u-1} B(F(x_u)) = 2^{-u} B(x_u).$$

The representation (76) is the definition with $u = \log_2[-\Theta(x)]$. ☐

Remark 7.3. The repository's sharp endpoint expansion contains a nonconstant periodic function Ψ of a lower-Lambert phase. The periodic factor $\mathcal{P}_{\text{dyn}}$ is different in construction: it is extracted from composition orbits and the exact global Schröder coordinate. Determining the transform between the two periodic objects is one of the main open problems proposed below.

8 Numerical experiments and exact symbolic checks

Role of computation. All theorems above are proved independently of numerics. The included Python program uses exact rational arithmetic for cumulants, moments, and Legendre coefficients, and common-random- number Monte Carlo only as a consistency check of the pathwise derivative formula.

8.1 Reproducibility package

The archive contains

- `code/fabius_q_response_experiments.py`, with detailed comments;
- exact CSV tables under `data/`;
- Monte Carlo validation tables under `data/`;
- the figures under `figures/`; and
- `NUMERICAL_README.txt` with the run parameters.

The reported run used 10^6 pathwise samples, 1.5×10^6 common-random-number CDF samples, 32 geometric digits, and a central-difference step $h = 0.005$. The omitted tail weight near $q = 1/2$ is approximately 2^{-32}, far below the sampling resolution.

8.2 Pathwise moment checks

For even m, theorem 4.1 gives

$$\dot{\mu}_m = m\mathbb{E}[Z^{m-1} V]. \tag{77}$$

The table compares exact values with the Monte Carlo estimator. The quoted uncertainty is one standard error.

order	exact $\dot{\mu}_m$	estimate	standard error
2	$-7.4074074 \times 10^{-2}$	-7.400607×10^{-2}	8.98×10^{-5}
4	$-8.1975309 \times 10^{-3}$	-8.188966×10^{-3}	1.54×10^{-5}
6	$-9.7019760 \times 10^{-4}$	-9.693966×10^{-4}	2.61×10^{-6}
8	$-1.2465992 \times 10^{-4}$	-1.246502×10^{-4}	4.52×10^{-7}
10	$-1.7132447 \times 10^{-5}$	-1.714973×10^{-5}	8.07×10^{-8}
12	$-2.4860408 \times 10^{-6}$	-2.491708×10^{-6}	1.48×10^{-8}
14	$-3.7717453 \times 10^{-7}$	-3.785464×10^{-7}	2.78×10^{-9}

All discrepancies are below one standard error in this run.

8.3 Tangent CDF and antisymmetry

A common set of digits was evaluated at $q = 1/2 + h$ and $q = 1/2 - h$, and the two empirical CDFs were differenced. The exact theory predicts $G(1-x) = -G(x)$. Across the displayed grid the root-mean-square antisymmetry residual was approximately 5.8×10^{-3}, compared with a peak response magnitude about 0.45.

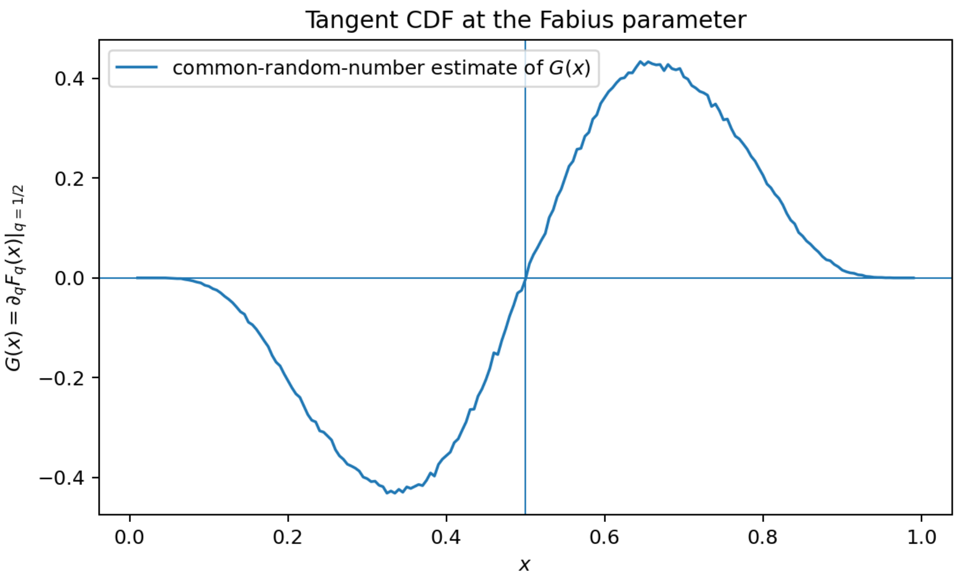

Figure 2: Common-random-number estimate of the tangent CDF. The exact function is antisymmetric about $x = 1/2$; the visible small mismatch is sampling noise and central-difference error.

9 Further deductions and frontier conjectures

9.1 Uniform transseries at the flat fronts

The exact edge defect (11) transports the small-argument asymptotic of F_q into a two-variable boundary layer. At a left-front point (q_0, x_0) with $q_0 = x_0 < 1/2$, put $q = x_0 + \epsilon$. Then

$$\Delta(q, x_0) = \frac{1}{1-q} F_q\left(\frac{\epsilon}{q}\right).$$

The flatness theorem says this is smaller than every power of ϵ. General-base atomic-function results in the repository strongly suggest the following sharper form.

Conjecture 9.1 (Uniform Lambert–periodic front expansion). *For every compact $K \Subset (0, 1/2]$, the endpoint expansion of $F_q(y)$ admits a version uniform for $q \in K$, including a lower-Lambert phase and a nonconstant smooth periodic correction. Consequently, as $\epsilon \downarrow 0$,*

$$\log \Delta(q_0 + \epsilon, q_0) \sim -\frac{\log^2(1/\epsilon)}{2\log(1/q_0)}, \tag{78}$$

with a complete expansion in logarithms, inverse logarithms, and a periodic phase whose coefficients vary smoothly with q_0.

A proof would turn the geometric plateau boundary into a laboratory for beyond-all-orders parameter asymptotics. The dyadic point is especially interesting because two fronts meet there and the defect acquires the factor two in (12).

9.2 Analyticity above the critical ratio

Conjecture 9.2 (Analytic supercritical parameter regime). *The map*

$$q \longmapsto f_q$$

is real analytic as a map into $C^\infty(\mathbb{R})$ on $(1/2, 1)$. On $(0, 1/2]$, the flat fronts of theorem 3.4 are the complete obstruction to such function-space analyticity.

The Fourier product is termwise analytic in q, but complexifying q destroys the simple real-frequency bounds because sinc factors can grow exponentially in imaginary directions. A successful proof will likely use a transfer-operator scale adapted to the overlapping regime, rather than a naive complex neighborhood of the Fourier product.

9.3 Asymptotics of the tangent Legendre spectrum

Problem 9.3 (Legendre phase and sign changes). *Determine the sharp asymptotic of a_{2n} in (45). In particular:*

1. *Does the sequence have infinitely many sign changes and infinitely many same-sign adjacent pairs?*
2. *Is its leading phase inherited from the endpoint Lambert-periodic correction, from the real-frequency shell profile of the sinc product, or from a new stationary phase?*
3. *Can one express its asymptotic through a saddle point involving the logarithmic derivative in (37)?*

The exact degree-16 sign failure is useful here: it rules out the simplest total-positivity ansatz and forces any asymptotic argument to retain a phase.

9.4 The dynamical periodic factor

Conjecture 9.4 (Nonconstant dynamical cocycle). *The function $\mathcal{P}_{\mathrm{dyn}}$ in (75) is nonconstant and real analytic. Its Fourier coefficients can be obtained by an explicit cohomological transform of the Gamma–zeta Fourier coefficients of the repository's endpoint periodic function Ψ.*

One route is to insert the sharp endpoint expansion into the cocycle $z(F(x)) - 2z(x)$, sum it along the orbit as in (70), and then convert the orbit phase to the global coordinate $u = \log_2[-\Theta(x)]$. The resulting Fourier transform should contain small divisors of the simple form $2^{1-2\pi i k/\log 2} - 1$, but the exact normalization depends on the lower-Lambert displacement.

9.5 A q-family of global dynamical coordinates

For $q < 1/2$, the density is exactly constant near the midpoint, so F_q itself is exactly affine there with multiplier $(1-q)^{-1}$. At $q = 1/2$, exact local affinity collapses to an affine Taylor jet plus a flat defect. For $q > 1/2$, the midpoint is no longer inside a plateau.

Problem 9.5 (Parameter-dependent Schröder coordinates). *Construct and analyze normalized coordinates Θ_q satisfying*

$$\Theta_q(F_q(x)) = \lambda_q \Theta_q(x), \qquad \lambda_q = f_q(1/2), \qquad \Theta_q'(1/2) = 1. \tag{79}$$

Determine the regularity of $(q, x) \mapsto \Theta_q(x)$ across $q = 1/2$. The front theorem predicts a C^∞, nonanalytic transition in the multiplier and potentially in the coordinate itself.

For $q < 1/2$, the local coordinate can be chosen literally linear throughout the plateau. At the critical dyadic ratio, it is linear only to infinite order. This makes $q = 1/2$ a natural bifurcation point between exact and asymptotic linearization.

9.6 Arithmetic of higher response jets

Conjecture 9.6 (Cyclotomic denominator sharpness). *After removing the von Staudt–Clausen denominator of B_{2m}, the odd denominator of $\partial_q^s \kappa_{2m}(1/2)$ contains the full cyclotomic contribution predicted by $(2^{2m}-1)^{s+1}$, except for cancellations governed by explicit divisibility of a universal derivative polynomial. The corresponding moment and Legendre jets admit a finite prime-by-prime valuation recursion through complete Bell polynomials.*

This would extend the repository's q-integer and Gaussian-binomial arithmetic from static moments to parameter response.

9.7 Thue–Morse finite models for susceptibility

The existing finite spline approximants have highest derivatives equal to finite Thue–Morse words. Differentiating their geometric weights with respect to q marks a digit position. This suggests a finite combinatorial model for ρ.

Problem 9.7 (Marked Thue–Morse spline approximation). *Construct finite signed splines ρ_N by differentiating the N-digit geometric-uniform box spline at $q = 1/2$. Prove:*

1. *an exact truncated-power formula whose coefficients are Thue–Morse signs weighted by a binary-position statistic;*
2. *sharp C^r convergence rates to ρ;*
3. *a finite-product Richardson acceleration in the quarter-base $q = 1/4$ scale; and*
4. *exact formulas for the top derivative plateaux and their signed value distribution.*

This program would unite the response theory with the repository's strongest exact finite models rather than treating ρ only through infinite products.

10 Lean formalization roadmap

The new results divide naturally into formalization tranches.

10.1 Tranche A: parameter calculus for the probability law

1. Prove uniform summability of $\partial_q^r[(1-q)q^j]$ on compact parameter intervals.
2. Define the coupled random variable $X(q,\omega)$ and its iterated parameter derivatives as uniformly convergent series.
3. Establish the pathwise test-function formula $\partial_q \mathbb{E}\varphi(X_q) = \mathbb{E}[\varphi'(X_q)\partial_q X_q]$.
4. Differentiate the affine independent-copy fixed point and package theorem 4.2 in the zero-mass Kantorovich–Rubinstein space.

The repository already has the affine independent-copy operator and uniqueness theorem, so the new work is principally differentiability and the signed-measure norm.

10.2 Tranche B: joint smoothness and flat fronts

1. Extend the geometric sinc factorization with uniform compact-parameter decay estimates.
2. Obtain joint C^∞ density regularity by Fourier inversion.
3. Formalize the exact plateau and edge-defect identities.
4. Prove that a jointly smooth function vanishing on one side of a smooth boundary has all mixed derivatives zero on the boundary.
5. Combine strict positivity of the edge CDF with the analytic identity theorem to prove nonanalyticity.

The front-jet formulas (15)–(17) should then be short corollaries.

10.3 Tranche C: exact response algebra

1. Differentiate the CDF integral equation and verify (32)–(34).
2. Add the marked sinc-product identity.
3. Reuse the existing Bernoulli cumulant and complete Bell infrastructure for (40) and (41).
4. Define shifted-Legendre response coefficients and prove (46) by polynomial coefficient extraction.

10.4 Tranche D: smooth dynamics

1. Derive the diagonal inequalities from strict convexity and prove global attraction of the inverse midpoint.
2. Formalize the flat contraction lemma theorem 6.3, first in a local one-dimensional real setting.
3. Globalize by finite inverse iteration and prove image $\mathbb{R}$, uniqueness, reflection, and the limit formula.
4. Deduce exact iterate conjugacy and the beyond-all-orders estimate.
5. Convert the endpoint quadratic law into the cocycle series (70) and the double-exponential orbit asymptotic.

The local linearization is the only substantial general-purpose analytic component. It is independent of Fabius-specific arithmetic and may be useful elsewhere in the library.

11 Conclusions

The geometric-uniform family contains two distinct kinds of smooth nonanalyticity. In the space variable, the classical Fabius function is smooth and nonanalytic because dyadic self-similarity transports endpoint flatness throughout its structure. In the parameter variable, the same endpoint flatness is pulled across the moving plateau boundary, producing a whole curve of smooth nonanalytic fronts. The critical dyadic point $(q, x) = (1/2, 1/2)$ is the meeting point of these mechanisms.

The tangent measure ν makes that critical point computable. It has a contractive susceptibility expansion, an exact refinement system, a marked infinite-sinc product, closed Bernoulli cumulants, Bell moment recurrences, and a rational Legendre spectrum. It is therefore a natural object for extending the repository's exact arithmetic and finite Thue–Morse approximants into parameter space.

The global coordinate Θ reveals an unexpectedly simple composition law beneath the nonanalytic graph: the whole open unit interval is smoothly conjugate to the real line under multiplication by two. The price of that simplicity is a nonanalytic, flat coordinate at the midpoint and extremely singular endpoint geometry. The logarithmic height B captures the latter, and its exact periodic quotient with Θ gives a concrete target for connecting composition dynamics to the known Lambert–periodic endpoint expansion.

These results suggest a coherent next phase: formalize parameter response, construct marked Thue–Morse finite models, solve the tangent Legendre asymptotics, and identify the Fourier content of the dynamical periodic factor.

A Derivation of selected exact coefficients

For illustration, $P_2(y) = (3y^2 - 1)/2$. Since the tangent measure has zero mass,

$$a_2 = 5 \cdot \frac{3}{2} \cdot 4\dot{\mu}_2 = 30\left(-\frac{2}{27}\right) = -\frac{20}{9}.$$

For $P_4(y) = (35y^4 - 30y^2 + 3)/8$,

$$\begin{aligned} a_4 &= 9\,(70\dot{\mu}_4 - 15\dot{\mu}_2) \\ &= 9\left[-70\left(\frac{83}{10125}\right) + 15\left(\frac{2}{27}\right)\right] = \frac{1088}{225}. \end{aligned}$$

Higher coefficients are obtained by the same triangular formula. The included script keeps all operations in exact rational arithmetic.

B Numerical file inventory

File	Contents
code/fabius_q_response_experiments.py	Exact symbolic tables, pathwise Monte Carlo checks, common-random-number CDF experiment, and figure generation.
data/exact_cumulant_and_moment_jets.csv	Exact $\kappa_n, \dot{\kappa}_n, \mu_n, \dot{\mu}_n$ through the configured order.
data/exact_tangent_legendre_coefficients.csv	Exact shifted-Legendre coefficients through degree 20.
data/monte_carlo_moment_derivative_validation.csv	Estimates, standard errors, and z-scores for $\dot{\mu}_{2m}$.

File	Contents
data/monte_carlo_legendre_validation.csv	Pathwise estimates and standard errors for the exact Legendre coefficients.
data/tangent_cdf_common_random_numbers.csv	Grid values of the central-difference estimate of G and its antisymmetry residual.
figures/tangent_cdf.png	Plot used in figure 2.
figures/tangent_legendre_coefficients.png	Plot used in figure 1.
NUMERICAL_README.txt	Reproduction parameters and a concise explanation of the numerical layer.

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